

Ahmed Fathy

Software Engineer

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EDUCATION

- **Cairo University, Faculty of Engineering** Giza, Egypt
B.Sc. in Computer Engineering; GPA: 3.7 2022 – 2027

EXPERIENCE

- **Al Nada Scientific Office** | *Technical Support Trainee* Cairo, Egypt | June 2022 – 2024
 - **Diagnostics & Maintenance:** Troubleshoot workstations and network equipment; performed firmware updates and preventive maintenance.
 - **Documentation:** Created systematic troubleshooting documentation to streamline hardware issue resolution.

PROJECTS

- **5-Stage Pipelined Processor (RISC)** | *VHDL, QuestaSim, Computer Architecture* 2025
 - **Core Architecture:** Engineered a 32-bit 5-stage pipelined processor in VHDL, supporting a custom ISA of over 25 instructions.
 - **Hazard Resolution:** Optimized throughput by implementing Hazard Detection and Data Forwarding units to resolve control and data dependencies.
 - **Verification:** Validated functional correctness through rigorous simulation and testbench environments in QuestaSim.
 - **CNN Convolution Accelerator** | *Verilog, OpenLane2, ASIC Design* 2025
 - **Architecture:** Designed a synthesizable 2D convolution accelerator using Systolic Arrays and fully pipelined dataflow to maximize MAC throughput.
 - **Memory Hierarchy:** Implemented DMA-based loading and dual-port SRAM buffering to support up to 16x16 kernel execution.
 - **Physical Design:** Executed RTL-to-GDSII synthesis using OpenLane2, validating physical feasibility and achieving timing closure.
 - **AES Encryption/Decryption Engine** | *Verilog, FPGA, Cryptography* 2023
 - **Core Logic:** Synthesized robust AES cryptographic core in Verilog HDL supporting 128-bit, 192-bit, and 256-bit key standards.
 - **Datapath:** Engineered hardware-optimized modules for SubBytes, ShiftRows, MixColumns, and AddRoundKey transformations.
 - **Validation:** Verified cryptographic correctness against standard NIST test vectors using rigorous debug tracing.
 - **Tactic (Robotics & AI)** | *Embedded Systems* 2024
 - **System Integration:** Developed browser-controlled robotics with AI vision and low-latency control
 - **Firmware:** Programmed ESP32 microcontrollers to handle WebSocket communication for real-time motor control.
 - **Backend Processing:** Deployed YOLOv5 and OpenCV on host server to process visual feedback and automate robotic decision-making.
 - **Hankers (Social Platform)** | *Full-Stack Engineering* 2025
 - **Architecture:** Built scalable social platform using Microservices principles, React.js, and TypeScript.
 - **Frontend:** Implemented responsive UI design and asynchronous API integration via AJAX.
 - **DevOps:** Established CI/CD pipelines and containerized services using Docker for consistent deployment.
- ## STUDENT ACTIVITIES
- **IEEE** | *UI/UX Designer - Frontend Team Leader* Cairo | Oct 2025 – Present
 - **Portfolio Platform Development:** Engineered comprehensive web platform featuring workshop and event management systems with admin dashboard for content creation, role-based access control enabling instructors to upload course materials and students to register for workshops and events.
 - **Technical Center for Career Development (TCCD)** | *Frontend Developer* Cairo | June 2025 – Present
 - **Portfolio Platform Development:** Built an integrated portfolio, event management, HR, and judging system with role-based access, dashboards, analytics, attendance tracking, custom forms, and evaluation workflows to streamline operations and decision-making.
 - **Physics Helping District (PhD)** | *Team Leader* Cairo | June 2022 – 2024
 - **Mentorship:** Led peer tutoring initiative in electronics, achieving measurable academic improvement.

- **NASA Space Apps Challenge** | *Participant – Project: "Orrery"*

Cairo | Oct 2024

- **Orrery App:** Developed an interactive orrery app visualizing near-Earth objects using React and Three.js.

TECHNICAL SKILLS

- **Digital Design & Verification:** Verilog HDL, VHDL, RTL Design, RISC-V ISA, MIPS Architecture, Pipeline Design, RTL-to-GDSII Flow, OpenLane2, DRC/LVS Verification, Static Timing Analysis, FSM Design
- **Development Tools & Platforms:** ModelSim, QuestaSim, Intel Quartus Prime, GTKWave, ESP32, AVR/Arduino
- **Programming Languages:** C/C++ (Advanced), Python, x86 Assembly, MIPS Assembly